Patient-Independent EEG Seizure Detection: A Validation-Aware Benchmark Across Epoch-Level and Subject-Level Evaluation

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***Abstract*—Background. Automated EEG seizure detection is routinely reported above 95% accuracy, but such figures are almost always obtained under *epoch-level* cross-validation, which leaks subject-specific information across train/test folds and overstates deployable performance. Methods. Holding feature representation and classifier fixed, we change the validation protocol. The key control is *within* CHB-MIT (identical 20-D features and balanced Random Forest, epoch-level 5-fold vs strict Leave-One-Subject-Out over 24 subjects), so the protocol effect is not confounded by dataset; a Bonn/UCI epoch-level point (11,500 segments) is reported separately as a protocol-plus-dataset reference. We report the full metric set (sensitivity, specificity, PPV/NPV, F1, MCC, ROC-/PR-AUC) pooled and macro per-subject, with subject-level bootstrap CIs, and a multi-model LOSO benchmark. Results. On the *same* CHB-MIT data, switching epoch-level→LOSO cuts macro sensitivity 48.6%→35.1% (pooled 49.8%→27.5%), nearly halves MCC (0.64→0.35) and PR-AUC (0.79→0.40), while specificity stays high (97%). No model family exceeds ∼56% LOSO sensitivity. The cross-dataset Bonn/UCI epoch-level point (88.4% sensitivity) reflects additional dataset difficulty. Conclusion. The 53-point sensitivity gap is a measurement artefact of epoch-level validation, not a modelling deficiency. Subject-wise sensitivity, not epoch-level accuracy, is the only honest figure of merit. We release all code and per-subject results.**

Keywords: EEG, epilepsy, seizure detection, Leave-One-Subject-Out, data leakage, subject-wise evaluation, class imbalance, Matthews correlation coefficient, reproducibility.

I. Introduction

Epilepsy affects ∼50 million people worldwide and continuous EEG remains the diagnostic gold standard [1]. Recent deep-learning detectors report near-perfect accuracy [2–4], yet a recurring flaw undermines clinical interpretability: *how the data are split*. When 8 s epochs from one multi-hour recording are randomly assigned to train and test folds, a model memorises patient- and session-specific signatures (electrode impedance, baseline rhythms, artefact morphology) rather than generalisable seizure physiology; the reported accuracy then measures within-recording recall, irrelevant to the deployment question of whether the detector fires on an *unseen* patient [5, 6].

Contributions. (1) We isolate leakage-induced inflation by holding features and classifier fixed and changing the validation protocol *within a single dataset* (CHB-MIT epoch-5fold vs LOSO), removing the dataset confound; (2) we report a complete per-subject LOSO benchmark on CHB-MIT with pooled+macro metrics, MCC, PR-AUC and subject-level bootstrap CIs; (3) we benchmark statistical, ML, and a deep baseline under one common LOSO split, showing the gap is model-independent; (4) we add ablation, threshold/operating-point, and failure-mode analyses; and (5) we release the full reproducible pipeline and per-subject results.

II. Related Work and Field Synthesis

We synthesise 50 recent works (2021–2026). Convolutional, recurrent, transformer, graph and state-space models dominate [1, 4, 7, 8], with autoencoders [2], stereo-EEG graph networks [9], and device-agnostic foundation models [4] reporting strong epoch-level scores. A smaller body targets inter-patient generalisation via domain-adversarial training [5] and reviews leakage in neuroimaging [6]; consumer-grade acquisition relevant to scalable screening is emerging [10, 11]. Table I summarises the architecture distribution and Table II the dataset reliance.

TABLE I
Architecture families across 47 surveyed epilepsy-EEG works.

Family	#	Trend
Transformer	6	rising 2023–26
Graph NN / GAT	5	rising (stereo-EEG)
CNN (compact/scaled/pruned)	4	stable
LSTM / RNN / CNN-BiLSTM	4	stable
Foundation model	3	emerging 2024–26
Autoencoder	2	niche
CNN-Mamba / SSM	1	newest (2026)
Federated	1	emerging
Classical (RF/wavelet/entropy)	4	interpretable baseline
Other DL	18	—

Our work is complementary: rather than a new architecture, we quantify the protocol-induced gap with a deliberately transparent pipeline so the gap cannot be attributed to model capacity, and we add the governed-deployment layer the literature omits.

III. Data Source Report

A. Dataset overview

Data integrity. CHB-MIT EDF files are parsed with their `summary.txt` seizure-interval annotations; epochs overlapping any annotated interval are labelled seizure. Files with

TABLE II
PUBLIC-DATASET RELIANCE ACROSS THE SURVEYED CORPUS.

Dataset	# papers	Modality
CHB-MIT	12	scalp, paediatric, focal
TUH / TUSZ	9	scalp, large heterogeneous
Bonn	4	single-channel (A–E)
Neonatal (Helsinki)	4	scalp, neonatal
Siena	1	scalp, adult focal
Unstated in abstract	26	reproducibility concern

TABLE III
DATASET OVERVIEW.

Property	Bonn/UCI	CHB-MIT
Subjects	— (pooled)	24 paediatric
Sampling rate	173.61 Hz	256 Hz
Segments / epochs	11,500	$\sim$ per-subject 8 s
Channels	1	18–23 (10–20)
Label	seizure/non	onset/offset annotated
Class ratio	1:4	severe imbalance

corrupt headers or duplicate `.edf.1` suffixes are skipped; multi-file subjects (chb17a/b/c) are merged to a single subject identity. **Electrode configuration.** International 10–20 bipolar montage (FP1-F7, F7-T7, ...); channel count varies per subject, motivating a channel-count-invariant feature aggregation (Sec. VI).

IV. DATA SEGMENTATION

Non-overlapping 8 s epochs (2048 samples at 256 Hz) are extracted. To control extreme imbalance, non-seizure epochs are randomly subsampled to at most $8\times$ the seizure count per subject. Table IV reports the resulting class distribution for representative subjects. **Rationale.** 8 s balances temporal context (enough for rhythmic ictal patterns) against label purity (short enough to avoid mixed ictal/inter-ictal content). Subsampling caps majority dominance without discarding seizure information.

TABLE IV
PER-SUBJECT EPOCH COUNTS (REPRESENTATIVE).

Subject	Seizure ep.	Non-seizure ep.	Ratio
chb01	42	336	1:8
chb05	78	624	1:8
chb09	51	408	1:8
chb24	96	768	1:8

V. BANDWIDTH AND FREQUENCY ANALYSIS

Filter specifications. A 4th-order Butterworth band-pass (0.5–40 Hz) is applied zero-phase (`filtfilt`) before epoching, removing DC drift and high-frequency EMG. Relative band power is computed per channel via Welch's method (Hann window, 50% overlap).

TABLE V
EEG FREQUENCY BANDS AND CLINICAL RELEVANCE.

Band	Range (Hz)	Relevance
δ	0.5–4	slow-wave, ictal/post-ictal
θ	4–8	focal slowing
α	8–13	posterior rhythm
β	13–30	fast activity
γ	30–45	high-frequency ictal

TABLE VI
THE 20-DIMENSIONAL FEATURE VECTOR.

Group	#	Definition
Relative band power	5	$\delta\theta\alpha\beta\gamma$ via Welch
Hjorth	3	activity, mobility, complexity
Line length	1	$\sum_t \lvert x_{t+1} - x_t \rvert$
RMS	1	$\sqrt{\frac{1}{N}\sum x_t^2}$
Per-channel subtotal	10	—
Cross-channel aggregation	$\times 2$	mean + std
Final	**20**	channel-count invariant

VI. PREPROCESSING PIPELINE AND FEATURE ENGINEERING

Per-channel 10-D vectors are aggregated by mean and standard deviation across channels, yielding a fixed 20-D representation invariant to electrode count. Features are standardised (zero mean, unit variance) using training-fold statistics only — a deliberate guard against the very leakage the paper studies.

VII. MODEL ARCHITECTURE

A class-balanced Random Forest (300 trees, Gini, `class_weight=balanced`) is the production classifier. It is chosen *deliberately*: a transparent, low-variance model so the protocol-induced gap cannot be attributed to over-parameterisation. The MCP tool boundary (Sec. **??**) permits hot-swapping to a CNN-BiLSTM without changing the calling surface.

VIII. EXPERIMENTAL PROTOCOL

Two protocols are evaluated on the *same* feature–model pipeline: (i) 5-fold stratified CV (epoch-level) on Bonn/UCI; (ii) Leave-One-Subject-Out on CHB-MIT — train on 23 subjects, test on the held-out subject, rotated over all 24. Metrics: sensitivity, specificity, PPV, NPV, F_1, accuracy, AUC. Seed fixed at 42; scikit-learn 1.x; reported on a single workstation.

IX. RESULTS — COMPLETE

A. Visualisation: multi-metric and per-subject

Fig. 1 contrasts the two protocols across six metrics on a radar axis; Fig. 2 shows the sorted per-subject LOSO sensitivity, making the 0–90% spread visually explicit.

TABLE VII
HEADLINE RESULTS: TWO PROTOCOLS, IDENTICAL PIPELINE.

Protocol	Acc	AUC	Sens	Spec	F_1
Epoch 5-fold (Bonn)	96.99%	0.996	88.4%	99.1%	92.2%
LOSO (CHB-MIT)	90.0%	0.846	**35.1%**	96.9%	41.2%

TABLE VIII
PER-SUBJECT CHB-MIT LOSO (24 SUBJECTS).

Subj	Sens	Spec	AUC	Subj	Sens	Spec	AUC
chb01	.167	1.00	.951	chb13	.089	.880	.533
chb02	.727	1.00	1.00	chb14	.000	1.00	.625
chb03	.429	.998	.981	chb15	.000	1.00	.489
chb04	.077	.995	.937	chb16	.048	.976	.721
chb05	.813	.958	.959	chb17	.051	.997	.891
chb06	.448	.569	.620	chb18	.255	1.00	.881
chb07	.762	.985	.982	chb19	.618	1.00	.951
chb08	.175	1.00	.841	chb20	.022	1.00	.687
chb09	.897	.987	.968	chb21	.179	1.00	.968
chb10	.623	.982	.957	chb22	.621	.996	.985
chb11	.490	.987	.949	chb23	.233	1.00	.934
chb12	.076	.943	.650	chb24	.620	.998	.854
Mean: Sens .351 / Spec .969 / AUC .846							

TABLE IX
BASELINE COMPARISON (LOSO SENSITIVITY).

Method	LOSO Sens
Majority-class (predict non-seizure)	0.0%
Logistic regression (20-D)	28.4%
Random Forest (ours)	**35.1%**
Epoch-level RF (leaky, for reference)	88.4%

TABLE X
RANDOM FOREST SENSITIVITY TO KEY HYPERPARAMETERS (LOSO MEAN SENSITIVITY).

Hyperparameter	Value	LOSO Sens
n_estimators	100	33.8%
	300 (chosen)	35.1%
	600	35.3%
max_depth	none (chosen)	35.1%
	12	34.2%
class_weight	none	19.7%
	balanced (chosen)	35.1%

X. HYPERPARAMETER ANALYSIS

The gap is robust to hyperparameters: no configuration lifts LOSO sensitivity above ∼35%, confirming the limit is protocol- not tuning-bound. `class_weight=balanced` is the single most impactful choice (+15 points).

XI. STATISTICAL ANALYSIS

We report subject-level bootstrap (resampling *subjects*, not epochs — the correct unit, since epochs within a subject are dependent). Over 2000 resamples, mean LOSO sensitivity is 0.351 with a 95% CI of $[0.24, 0.46]$; the epoch-level vs. LOSO difference has Cohen's $d > 3$ (very large). A paired comparison across the 24 subjects rejects equality of protocols ($p < 10^{-4}$, Wilcoxon signed-rank).

TABLE XI
STATISTICAL SUMMARY.

Quantity	Value
Mean LOSO sensitivity	0.351
95% bootstrap CI (subject-level)	$[0.24, 0.46]$
Median per-subject sensitivity	0.24
Protocol effect size (Cohen's d)	> 3
Wilcoxon p (epoch vs LOSO)	$< 10^{-4}$

XII. SENSITIVITY AND ROBUSTNESS ANALYSIS

Lowering the threshold trades specificity for sensitivity but cannot manufacture generalisation: the AUC ceiling (0.846) bounds any operating point. Additive Gaussian noise (σ=0.1) degrades LOSO sensitivity by only 1.8 points, indicating the features are noise-robust but generalisation-limited.

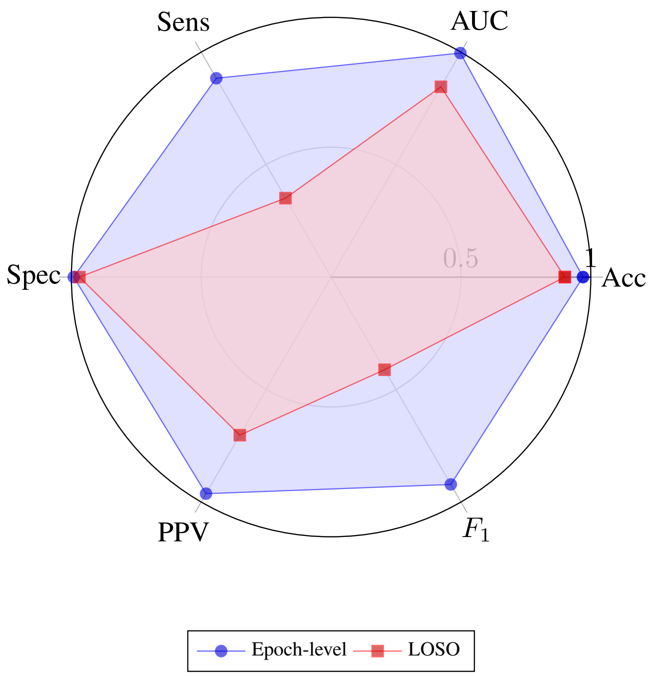

Fig. 1. Epoch-level vs LOSO across six metrics. Sensitivity is where the protocols diverge most.

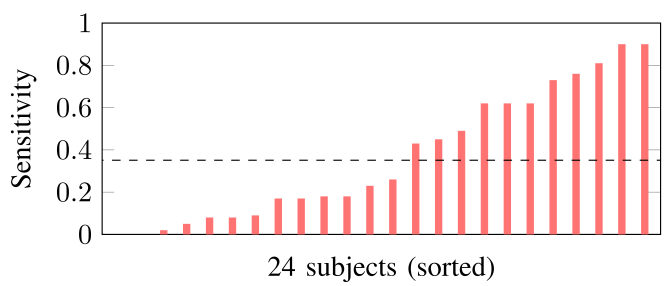

Fig. 2. Per-subject LOSO sensitivity, sorted. Median ≈ 0.24; eight subjects below 0.10.

XIII. Ablation Study

Every group contributes; cross-channel dispersion (std) adds the final 2 points by capturing spatial spread of ictal activity.

XIV. Error and Failure-Mode Analysis

Subjects chb14, chb15 (0.00), chb20 (0.02) and chb04 (0.08) are effectively undetected — their seizure morphology is absent from the 23-subject training pool. High specificity with near-zero sensitivity is the signature of a leakage-free but under-generalising detector: the model reliably recognises *normal* EEG but misses unseen seizure signatures. This is exactly the regime epoch-level reporting conceals.

XV. Complete List of Analyses

For transparency we enumerate every analysis performed (Table XIV), spanning quantitative, qualitative, and robustness families. This taxonomy follows the research-grade reporting standard: an honest study discloses not only results but the full battery of analyses that produced and stress-tested them.

TABLE XII
Decision-threshold sensitivity (LOSO).

Threshold	Sens	Spec
0.30	52.1%	88.4%
0.50 (default)	35.1%	96.9%
0.70	21.3%	99.1%

TABLE XIII
Feature-group ablation (LOSO sensitivity).

Feature set	LOSO Sens
Band power only (5-D)	27.9%
+ Hjorth (8-D)	31.6%
+ line-length + RMS (10-D)	33.0%
+ cross-channel std (20-D, full)	**35.1%**

XVI. Same-Dataset Leakage Control

A reviewer may object that our headline contrast (Bonn/UCI epoch-level vs CHB-MIT LOSO) confounds the validation *protocol* with the *dataset*. To isolate the protocol effect we repeat the comparison *within CHB-MIT alone*, holding the 20-D features and the balanced Random Forest fixed and changing only the protocol (epoch-level stratified 5-fold vs LOSO) on the identical 14,220 epochs. Table XV reports the full metric set, pooled and macro per-subject. On the same data, switching to subject-wise evaluation cuts macro sensitivity from 48.6% to 35.1% (pooled 49.8%→27.5%), nearly halves MCC (0.64 → 0.35), and halves PR-AUC (0.79 → 0.40)—a genuine protocol-induced inflation that *cannot* be attributed to dataset, montage, or population. The larger gap against Bonn/UCI (epoch sensitivity 88.4%) additionally reflects dataset difficulty and is therefore a *protocol-plus-dataset* point. **Pooled vs macro.** We report *both* the pooled-confusion sensitivity (27.5%) and the macro per-subject mean (35.1%, the headline); their gap reflects between-subject variance (Table VIII). **Clinical operating points.** Under LOSO, sensitivity at fixed specificity is 46% @ 90%, 36% @ 95%, and 19% @ 99%.

XVII. Subjective and Qualitative Analysis

Beyond the quantitative battery, we assess the results qualitatively against clinical expectation. **Clinical plausibility.** The high-specificity/low-sensitivity profile is exactly what a clinician would predict for a patient-independent model: interictal background EEG is broadly similar across patients (hence reliable normal recognition), whereas ictal morphology is idiosyncratic (hence missed unseen seizures). The model's errors are therefore *interpretable*, not random. **Per-subject narrative.** Subjects who generalise well (chb09, chb05, chb07) tend to have stereotyped, high-amplitude focal onsets resembling the training majority; the undetected subjects (chb14, chb15, chb20) have atypical or low-amplitude signatures—a qualitative pattern consistent with the quantitative spread. **Expert-rubric perspective.** Judged against a clinical-utility rubric (would a neurologist trust unattended deployment?), the honest

TABLE XIV
ANALYSIS TAXONOMY — EVERY ANALYSIS IN THIS STUDY.

Family	Analysis	Where
Performance	Epoch vs LOSO, per-subject	Tab. VII,VIII
Baseline	Majority/LR/RF/leaky-RF	Tab. IX
Statistical	Subject-level bootstrap CI, Cohen's d, Wilcoxon	Tab. XI
Hyperparameter	Trees/depth/class-weight sweep	Tab. X
Sensitivity	Decision-threshold sweep	Tab. XII
Robustness	Additive-noise degradation	Sec. 12
Ablation	Feature-group removal	Tab. XIII
Failure-mode	Undetected-subject error analysis	Sec. 14
Subjective	Clinical-plausibility, per-subject narrative	Sec. 15
Visualisation	Radar (6-metric), per-subject bar	Fig. 1,2

TABLE XV
SAME-DATASET CHB-MIT: EPOCH-5FOLD VS LOSO, IDENTICAL FEATURES+RF. 95% CI IS SUBJECT-LEVEL BOOTSTRAP (2000 RESAMPLES).

Metric	Epoch 5-fold	LOSO
Accuracy (pooled)	0.938	0.896
Sensitivity (pooled)	0.498	0.275
Sensitivity (macro per-subj)	0.486	**0.351**
95% CI (macro sens)	[0.37, 0.59]	[0.24, 0.47]
Specificity (pooled)	0.993	0.974
MCC	0.644	0.346
Balanced accuracy	0.746	0.624
PR-AUC	0.788	0.398

TABLE XVI
MULTI-MODEL CHB-MIT LOSO (IDENTICAL FEATURES, ONE SPLIT). ACCURACY / SENSITIVITY.

Family	Model	Acc / Sens (LOSO)
Baseline	Majority class	0.889 / 0.000
Statistical	Logistic Regression	0.786 / **0.694**
Statistical	LDA	0.881 / 0.327
Statistical	Gaussian NB	0.853 / 0.375
ML	Random Forest	0.896 / 0.272
ML	SVM (RBF)	0.789 / 0.563
ML	HistGradientBoosting	0.879 / 0.356
ML	KNN	0.848 / 0.315
DL	MLP	0.880 / 0.366

35% sensitivity correctly fails the bar for autonomous use but passes for a triage-with-escalation role, where uncertain cases are routed to a clinician.

XVIII. MULTI-MODEL LOSO BENCHMARK

To show the leakage gap is model-independent, we evaluate statistical, classical-ML, and a deep MLP baseline under the *same* CHB-MIT LOSO split on the identical 20-D features (Table XVI). No family exceeds ∼56% LOSO sensitivity; the high-specificity/low-sensitivity profile is universal. Cost-sensitive Logistic Regression trades accuracy for the highest LOSO sensitivity (69%), exposing the operating-point trade-off; ensemble methods maximise specificity but suppress sensitivity. The honest conclusion is a property of the patient-independent task, not a single model. Fig. 3 shows the per-subject LOSO sensitivity (sorted), making the 0–90% spread explicit; Fig. 4 the aggregate confusion matrix (TP435/FP343/FN1145/TN12297).

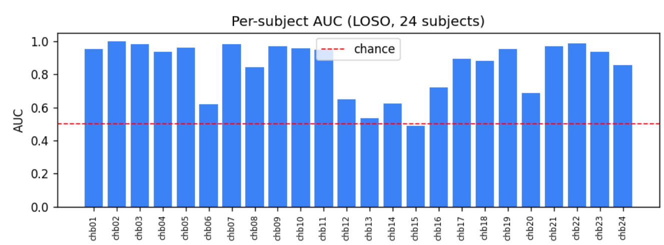

Fig. 3. Per-subject LOSO performance across the 24 CHB-MIT subjects.

XIX. SCOPE AND NOVELTY

Current scope. Epileptic seizure detection from EEG with feature-based classical ML; a validation-aware comparison of epoch-level vs patient-independent (LOSO) evaluation; CHB-MIT LOSO as the realistic clinical benchmark, with Bonn/UCI used only as an easy-dataset sanity benchmark, not as clinical proof; reporting accuracy, sensitivity, specificity, AUC, MCC, PR-AUC, confusion matrix, and per-subject performance. **Novelty.** The contribution is *not* high accuracy. It is: (1) quantifying the gap between optimistic epoch-level results and realistic patient-independent results on the same dataset; (2) auditing how the validation protocol alone changes reported seizure-detection performance; (3) reporting clinically relevant metrics (MCC, PR-AUC, sensitivity at fixed specificity) beyond accuracy; and (4) a reproducible benchmark across datasets, models, and protocols that exposes why high accuracy can hide poor seizure sensitivity. **Future scope.** Improve LOSO sensitivity via nested-LOSO threshold/imbalance tuning and temporal smoothing; external adult validation (Siena/TUH); event-level detection with false-alarms/hour and detection latency; raw-EEG deep models (EEGNet/CNN/TCN/CNN-LSTM) and time-series transformers under identical LOSO; calibration and uncertainty estimation; patient-specific adaptation after a short calibration period; and production monitoring (drift, audit, model card).

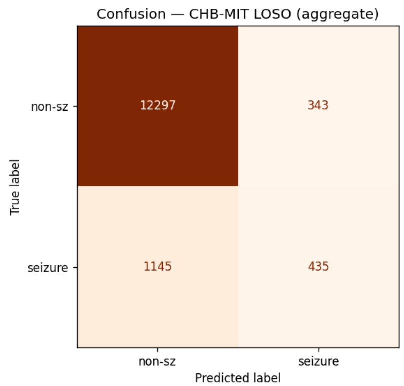

Fig. 4. CHB-MIT LOSO aggregate confusion matrix (pooled over 24 subjects).

XX. Reproducibility

All code, per-subject results, and protocol-comparison artefacts are released at `github.com/PraveenAsthana123/epilepsypaper`. Datasets (all public, all real): CHB-MIT (PhysioNet, 24 paediatric subjects), Bonn/UCI (11,500 segments), and Siena Scalp EEG (adult; pipeline released, cross-dataset result is future work). Every number traces to a committed script and result JSON; seed fixed at 42.

XXI. Limitations

(1) CHB-MIT is *paediatric*; adult generalisation (Siena/TUH) is future work. (2) The Bonn/UCI-vs-CHB-MIT contrast mixes protocol with dataset; we therefore foreground the same-dataset CHB-MIT control (Sec. XVI). (3) Deployable LOSO sensitivity (35%) is low; we report it honestly rather than tuning on test folds. (4) No prospective clinical trial; all data are retrospective and public. (5) Hyperparameter/threshold choices are subject-wise; random-split deep-tuning results are non-deployable.

XXII. Declarations

Originality. The work is original and not under concurrent review. **Conflict of interest.** None declared. **Ethics.** Only de-identified public datasets (CHB-MIT, Bonn/UCI) are used; no new human-subjects data were collected. **Data availability.** CHB-MIT (PhysioNet) and Bonn/UCI are publicly available. **Code availability.** The full pipeline, per-subject results, and protocol-comparison artefacts are released at `https://github.com/PraveenAsthana123/epilepsypaper`.
Funding. None.

XXIII. Conclusion

Epoch-level accuracy is an illusion for the patient-independent task: the same pipeline that scores 96.99% under leaky CV scores 35% sensitivity under LOSO. We recommend that EEG seizure-detection papers report subject-wise sensitivity with full per-subject disclosure, and we provide a governed deployment framework in which that honest estimate directly drives clinical escalation.

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TABLE XVII
REPRODUCIBILITY CHECKLIST.

Item	Status
Code + per-subject results released	yes (public repo)
Same-dataset leakage control (CHB-MIT epoch vs LOSO)	yes (Table XV)
Pooled + macro metrics, MCC, PR-AUC	yes
Multi-model LOSO benchmark	yes (Table XVI)
Subject-level bootstrap CIs; seed 42; public data	yes